# 전자공학실험 Term Project

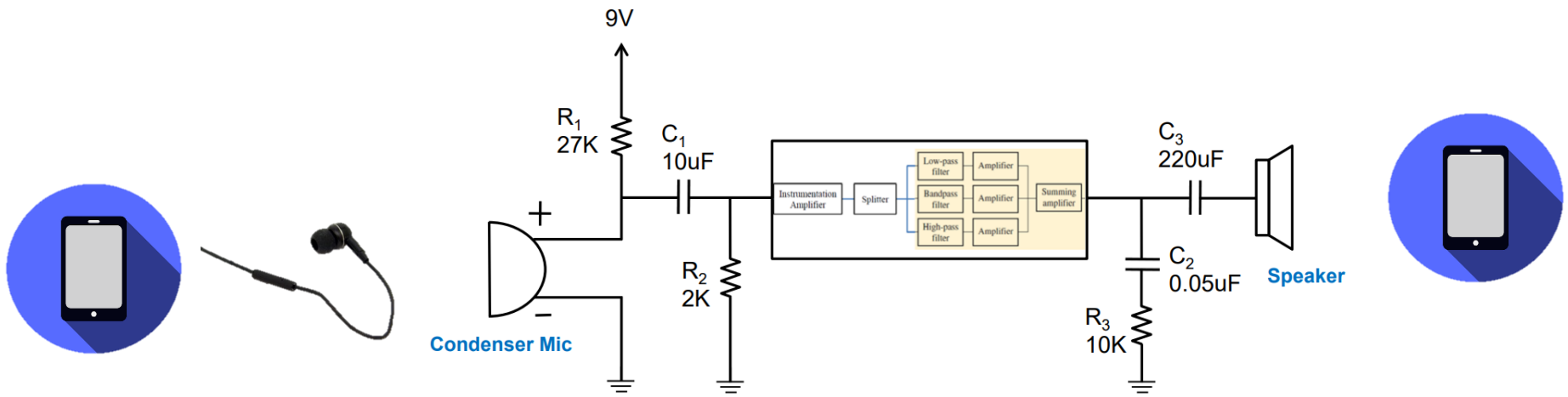
# Term project

- 목차 / Table of Contents


| 섹션 / Section | 한국어 | English |
|---|---|---|
| 실험 개요 / Overview | 3 | 13 |
| 신호 흐름 / Signal flow | 4 | 14 |
| 코드·음원 다운로드 / Download code and audio | 5 | 15 |
| Google Drive 업로드 / Upload to Google Drive | 6 | 16 |
| 녹음 파일 저장 / Save your recording | 7 | 17 |
| Colab 경로 설정 / Set the Colab path | 8–9 | 18–19 |
| 샘플·필터 선택 / Choose sample and filter | 10 | 20 |
| 결과 확인 / Read the results | 11 | 21 |
| 평가 지표 (SI-SNR) / Evaluation metric (SI-SNR) | 12 | 22 |

# Term project

- Sound equalizer
  - ✓ Condenser mic.에 음성(speech)/음악(music)을 핸드폰, 노트북 등으로 재생
  - ✓ Speaker에 핸드폰, 노트북 등 녹음이 가능한 기기를 이용하여 녹음
    - m4a로 저장되는 경우 https://convertio.co/kr/m4a-wav/ 또는 https://online-audio-converter.com/ko/ 등을 이용하여 wav 확장자명으로 저장
    - 최대한 조용한 환경에서 재생/녹음 진행
    - 가급적 재생/녹음 시작과 중지 모두 동시에 진행
    - 재생 음원이 녹음 장치에 녹음되지 않도록 주의

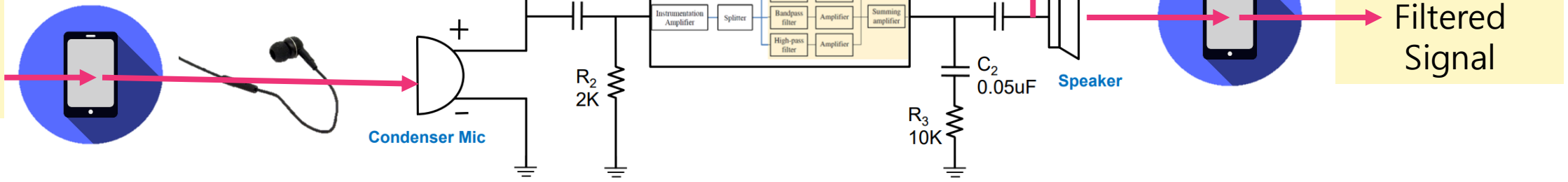



녹음 환경 예시

# Term project

- Source

Source Signal (재생음원)




Analog Filtered Signal

Recorded Filtered Signal

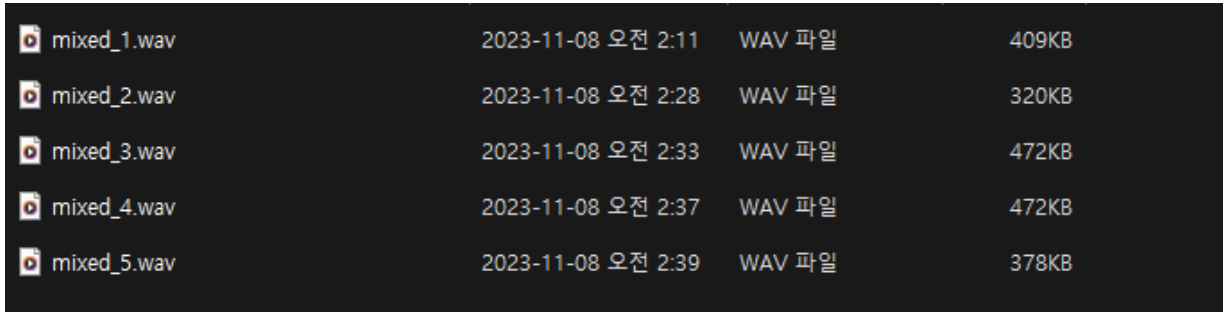
녹음 환경 예시

원음을 그대로 사용하거나
같은 환경에서 녹음해도 됨

Ideal Filtered Signal

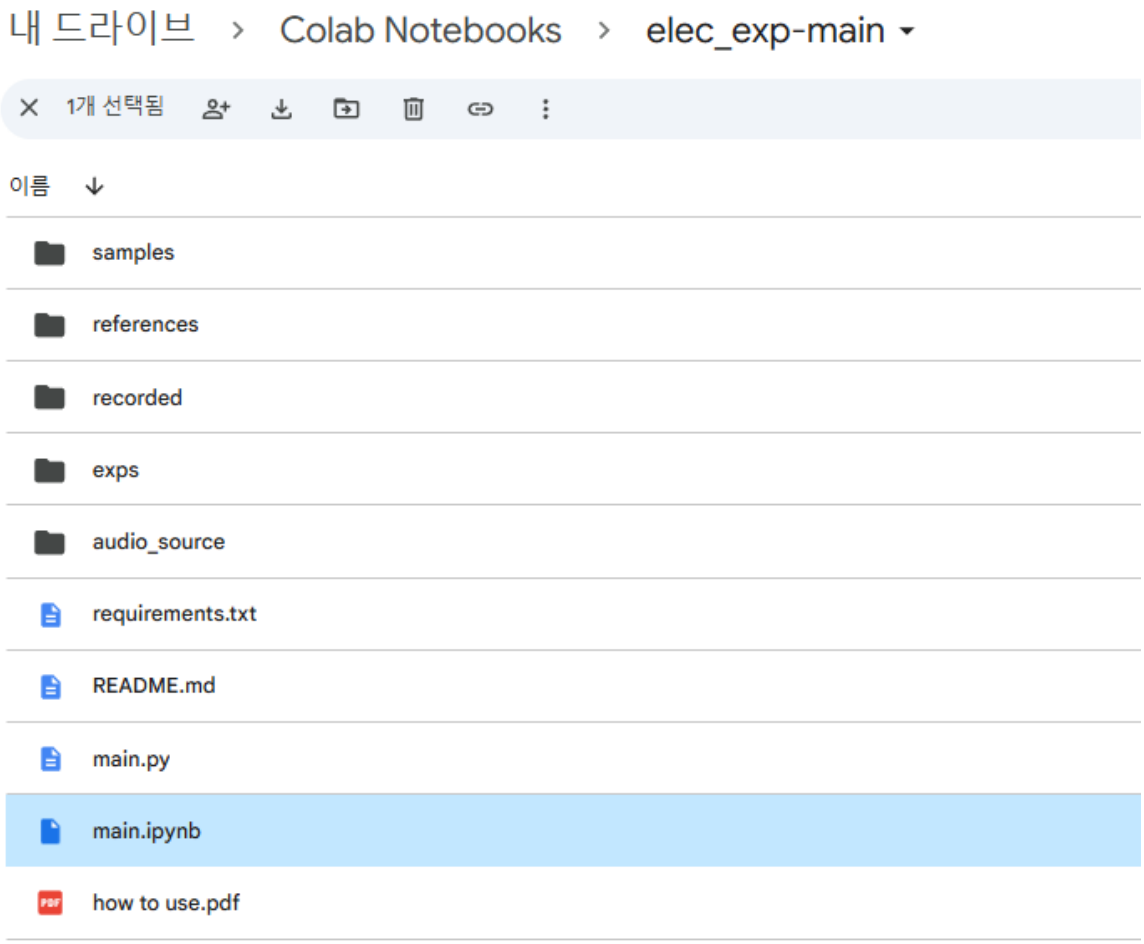
# Term project

- 실험 분석
  - ✓ 아래 github 주소에서 음원 및 스크립트 다운로드
    - https://github.com/9ooDa/26_elec_lab_term_proj
  - ✓ 실험 방법은 개인 PC에서 실행하는 방법과 Google Colab을 이용하여 온라인에서 진행하는 방법이 있음
    - Colab에서 진행하는 방법은 PC에 설치를 요구하는 것이 없음
    - 자세한 내용은 github의 readme.md 참조
  - ✓ 음원 샘플은 총 5개이며, samples폴더에 존재하고 파일 이름에 샘플 번호가 아래와 같이 표기
    - Ex) mixed_1.wav (1번 음원 샘플)



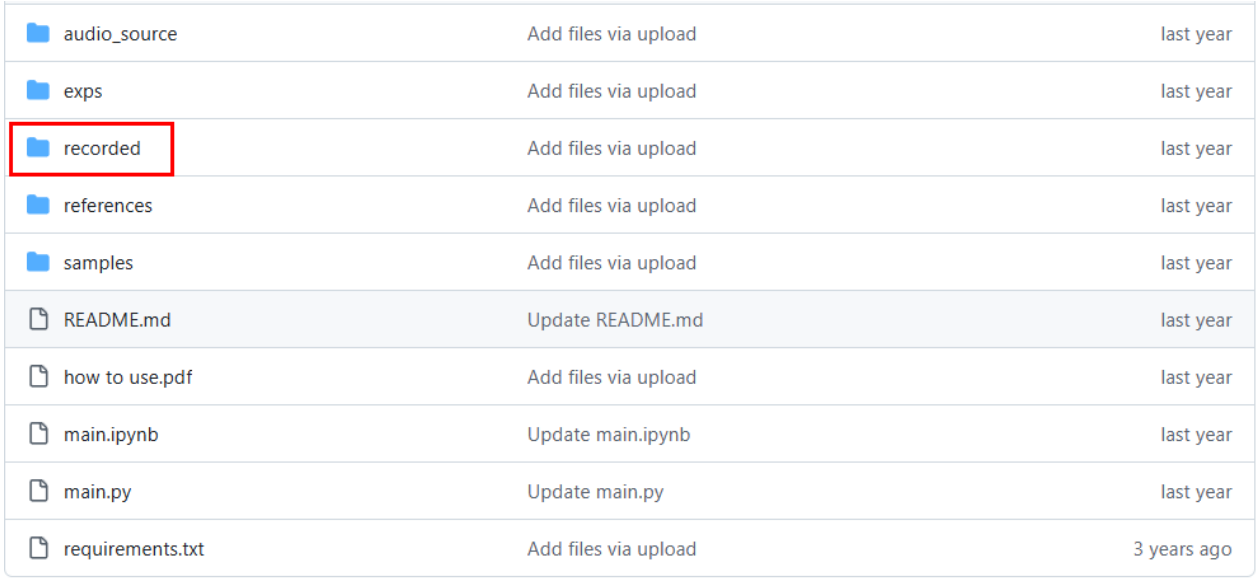
# Term project

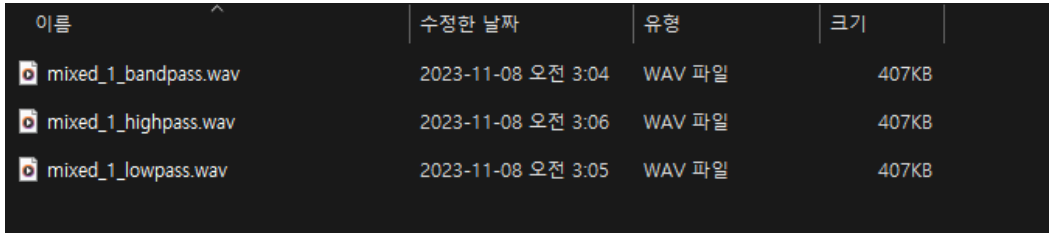
- Using Colab : Upload to your drive



# Term project

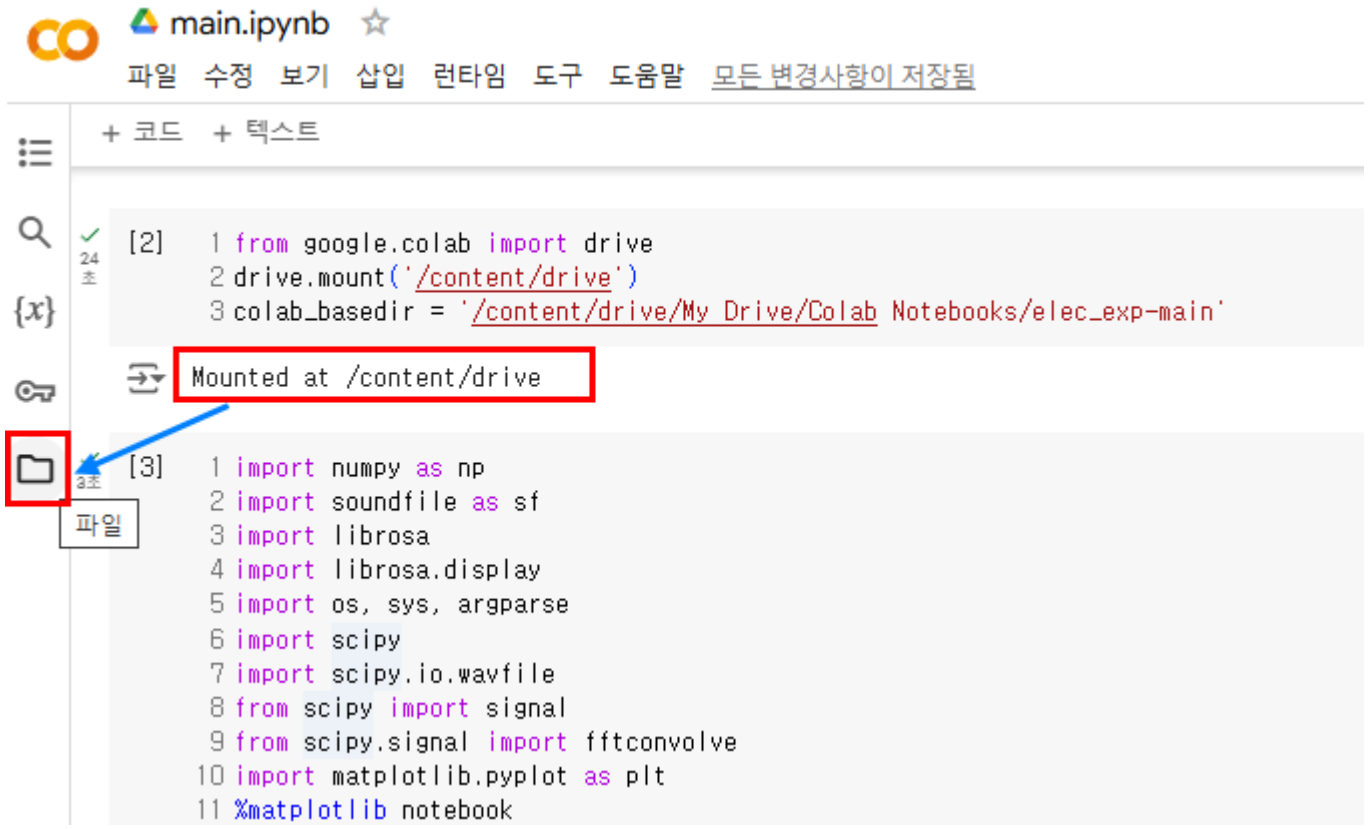
- 실험 분석
  - ✓ 녹음된 오디오 파일을 아래와 같은 경로에 저장
    - 설계한 회로에서 재생되는 음원을 wav 파일로 변환한 뒤, recorded 폴더에 **mixed_{sampld_idx}_{filter}.wav**로 저장
      - {sample_idx}에는 샘플 음원의 번호를, {filter}에는 실험하는 filter의 종류를 삽입
      - Ex) mixed_1.wav 파일 (음원 샘플 1번)으로 lowpass를 실험하는 경우 mixed_1_lowpass.wav



# Term project

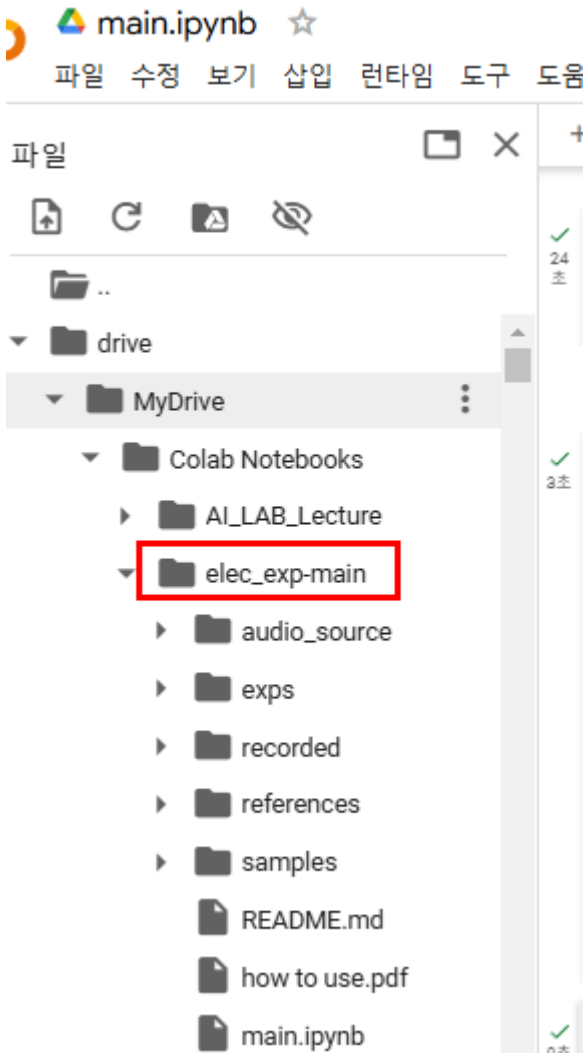
- 실험 분석




Check the real path

# Term project

- 실험 분석




```
1 from google.colab import drive
2 drive.mount('/content/drive')
3 colab_basedir = '/content/drive/My Drive/Colab Notebooks/elec_exp-main'
```

```
Mounted at /content/drive
```

colab_basedir must match the left folder path

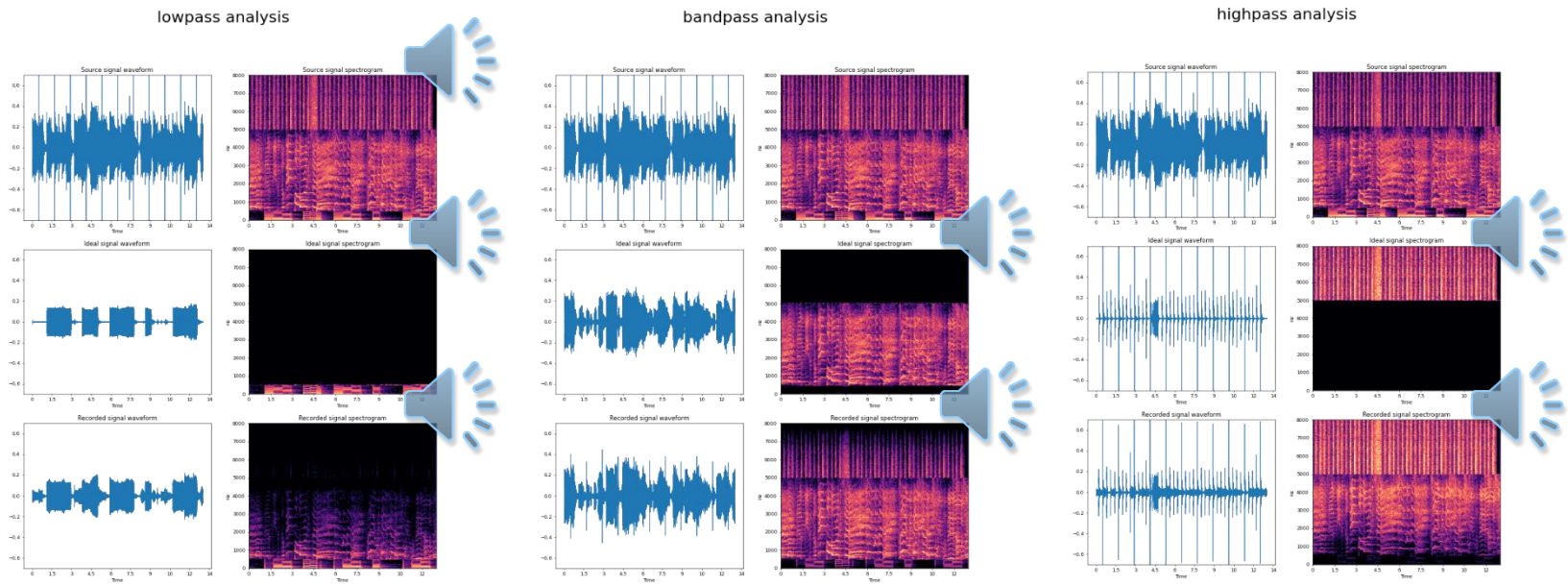
# Term project

- 실험 분석
  - ✓ main.py 또는 main.ipynb에서 'sample_idx' 와 'filter_mode' 변수만을 수정
    - 'sample_idx'는 음원 샘플의 index를 나타내는 것으로 1~5사이의 정수 값을 입력
    - Lowpass를 실험하는 경우 'filter_mode'를 'lowpass', highpass를 실험하는 경우 'highpass', bandpass를 실험하는 경우 'bandpass'로 설정하여 실험

```
[4]  # MODIFY THE SAMPLE AND FILTER
     sample_idx = 1 # 1 ~ 5
     filter_mode = 'lowpass' # lowpass, highpass, or bandpass
```

# Term project

- 실험 분석
  - ✓ 입력 신호(source signal), 이상적인 필터 통과 신호(ideal signal), designed 필터 통과 신호에 대한 waveform 및 spectrogram 비교
    - Spectrogram은 시간-주파수에 대한 표현으로 x축은 시간, y축은 주파수를 표기
    - 스크립트 실행 시, 우측 그림과 같은 그림 파일 생성
    - 제일 상단은 source signal, 가운데는 ideal signal, 제일 하단은 designed signal을 나타냄



# Term project

- 실험 분석
  - ✓ Signal-invariant signal-to-noise ratio (SI-SNR) 객관적 지표를 사용하여, ideal 신호와 designed 신호와의 유사도 평가
    - 처리 과정에서 발생된 왜곡된 신호에 대한 비율을 나타내는 객관적 지표

$$SI-SNR := 10\log_{10}\left(\frac{\|\boldsymbol{s}_{target}\|^2}{\|\boldsymbol{e}_{noise}\|^2}\right),$$

분모(오차)가 작을 수록 큰 수치를 나타냄

$$\boldsymbol{s}_{target} = \frac{\langle \hat{\boldsymbol{s}}, \boldsymbol{s}\rangle}{\|\boldsymbol{s}\|^2}\boldsymbol{s}$$

Ideal 신호를 designed 신호의 크기로 정규화

$$\boldsymbol{e}_{noise} = \hat{\boldsymbol{s}} - \boldsymbol{s}_{target}$$

정규화된 ideal 신호와 designed 신호와의 차
= filter 오차율

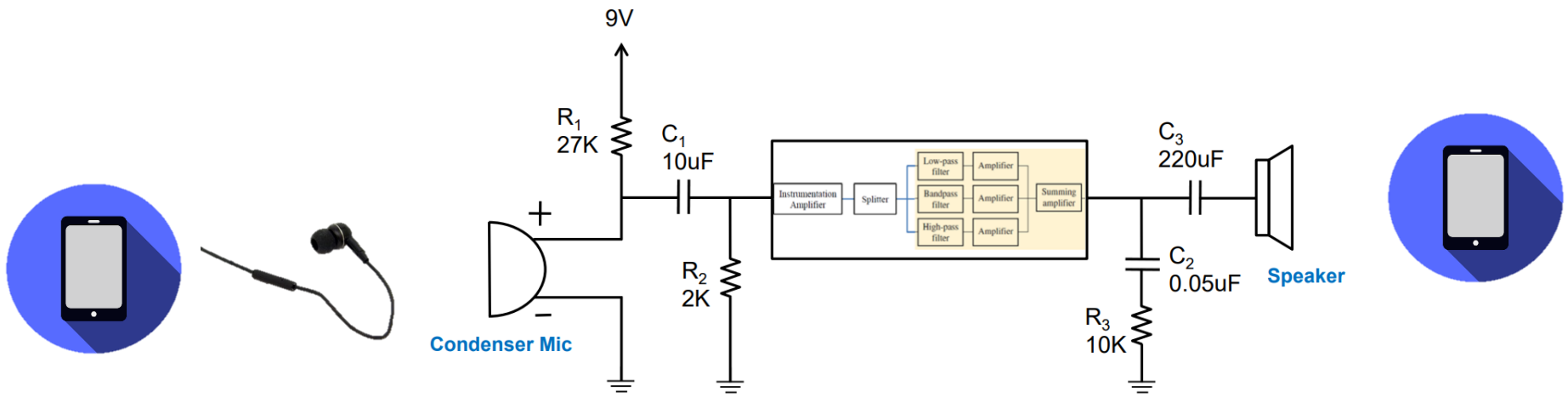
$where,\ S$: ideal signal (이상적인 신호),

$\hat{S}$: filtered signal (설계한 필터를 통과한 신호)

    - Designed 신호가 ideal 신호와 유사할 수록 높은 score를 나타냄
      - Second order butterworth digital filter를 이용한 경우 SI-SNR: 7.98dB
      - Analog filter로 설계된 경우 녹음 방법 등으로 인해 해당 수치보다 낮을 수 도 있음

# Term project

- Sound equalizer
  - ✓ Play speech/music into the condenser mic. with a phone, laptop, etc.
  - ✓ Record the speaker output with a device that can record (phone, laptop, etc.)
    - If saved as m4a, convert it to wav with https://convertio.co/kr/m4a-wav/ or https://online-audio-converter.com/ko/
    - Play and record in as quiet a place as possible
    - Start and stop playback and recording at the same time if possible
    - Make sure the original playback is not picked up directly by the recorder

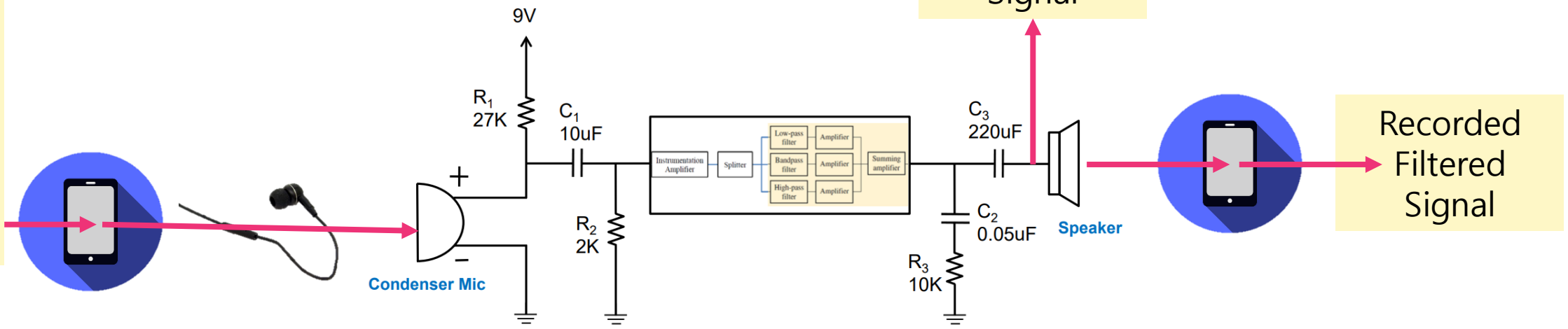



Example recording setup

# Term project

- Source

Source Signal (audio)




Analog Filtered Signal

Recorded Filtered Signal

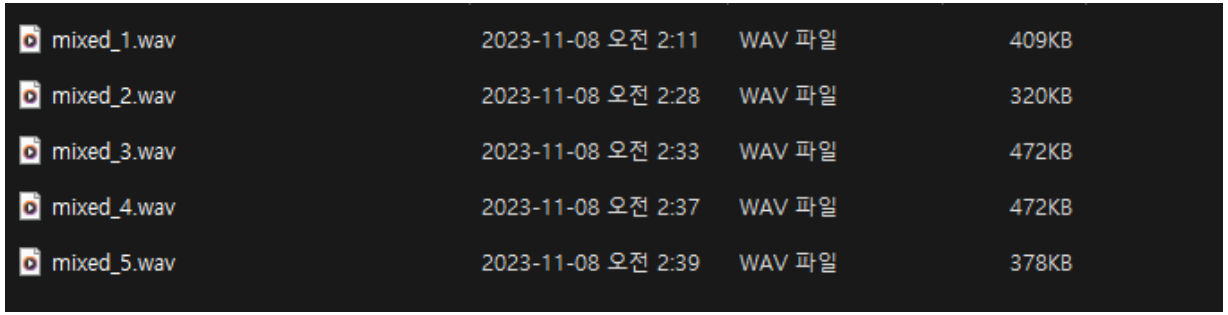
Example recording setup

Use the original audio as is, or record it in the same setup

Ideal Filtered Signal

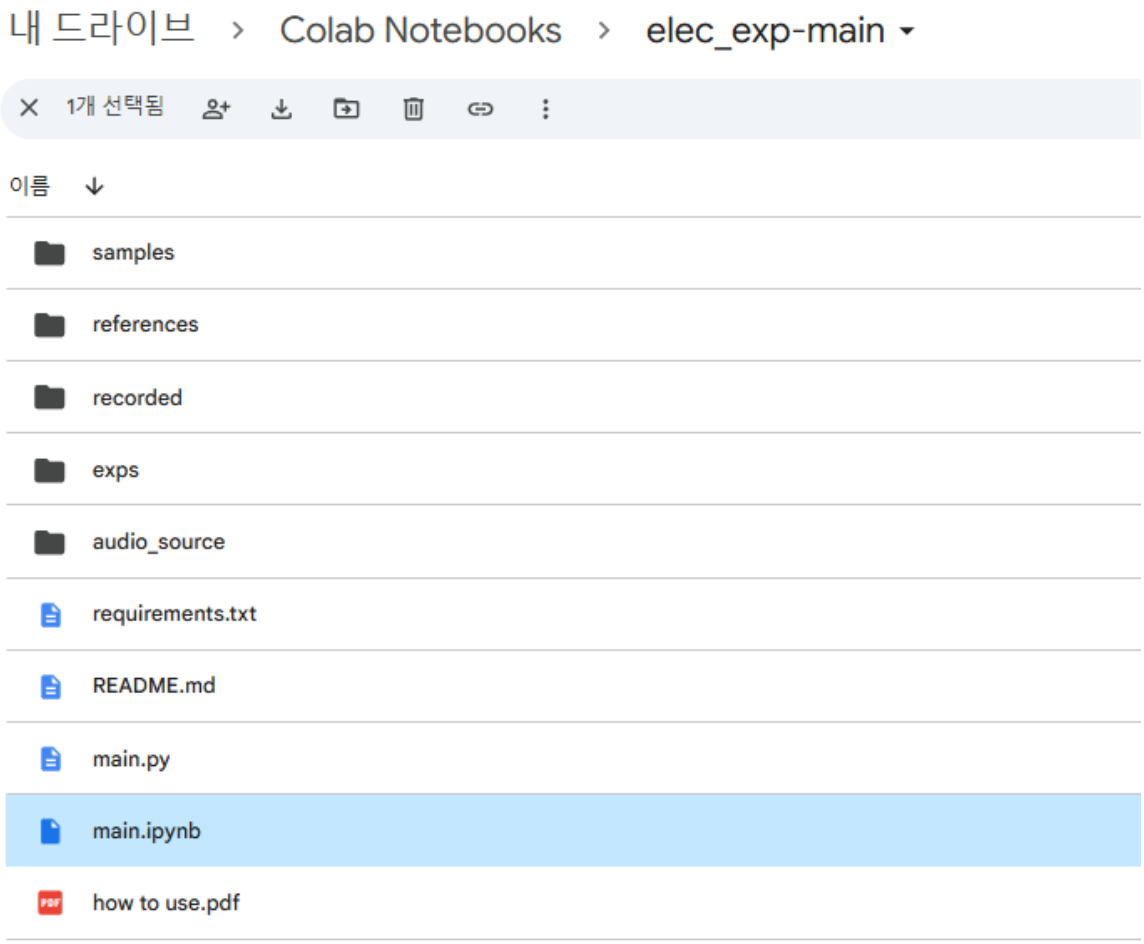
# Term project

- Experiment analysis
  - ✓ Download the audio files and scripts from the GitHub address below
    - https://github.com/9ooDa/26_elec_lab_term_proj
  - ✓ You can run the experiment on your own PC or online with Google Colab
    - Colab needs no installation on your PC
    - See README.md on GitHub for details
  - ✓ There are 5 audio samples in the samples folder; the file name shows the sample number as b elow
    - Ex) mixed_1.wav (audio sample 1)



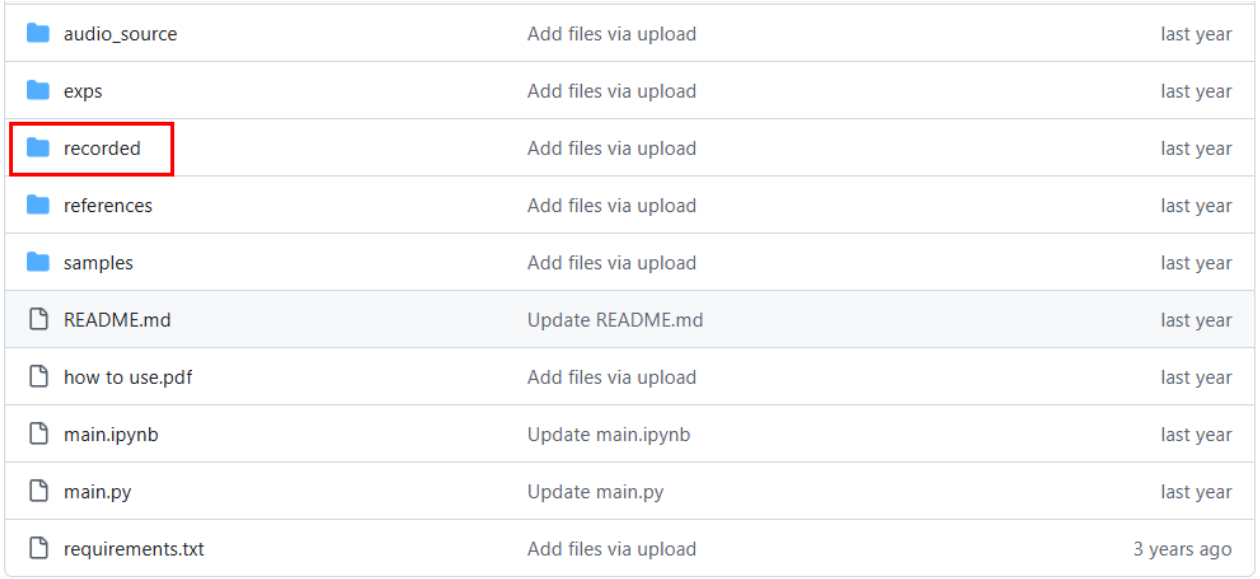
# Term project

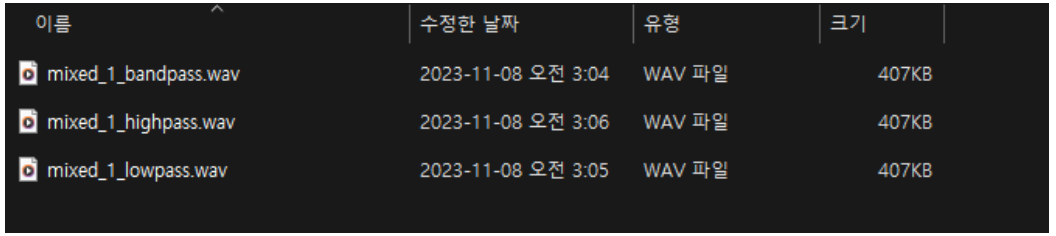
- Using Colab : Upload to your drive



# Term project

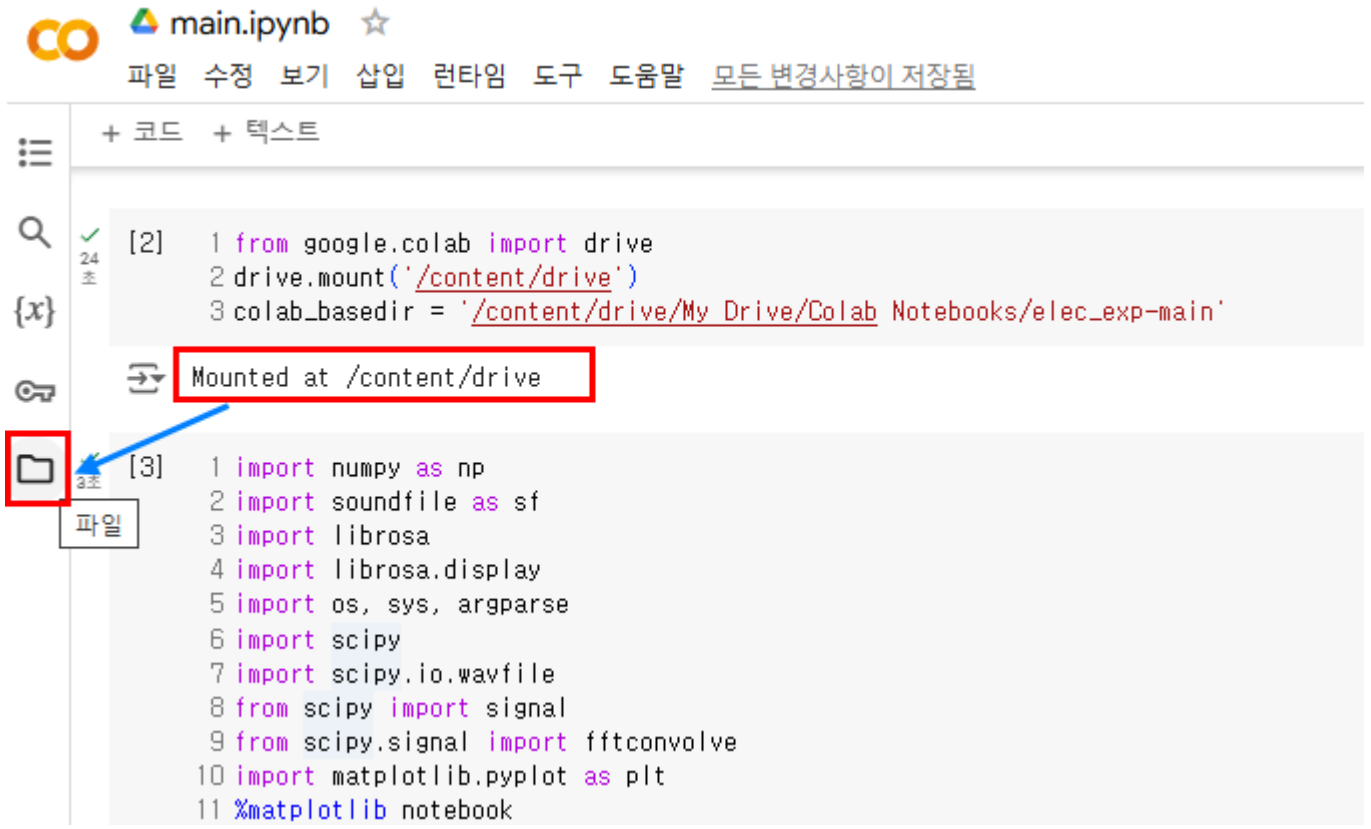
- Experiment analysis
  - ✓ Save the recorded audio file at the path below
    - Convert the audio played through your circuit to wav and save it in the recorded folder as **mixed_{sample_idx}_{filter}.wav**
      - Replace {sample_idx} with the sample number and {filter} with the filter type you tested
      - Ex) lowpass experiment with mixed_1.wav (audio sample 1) → mixed_1_lowpass.wav



# Term project

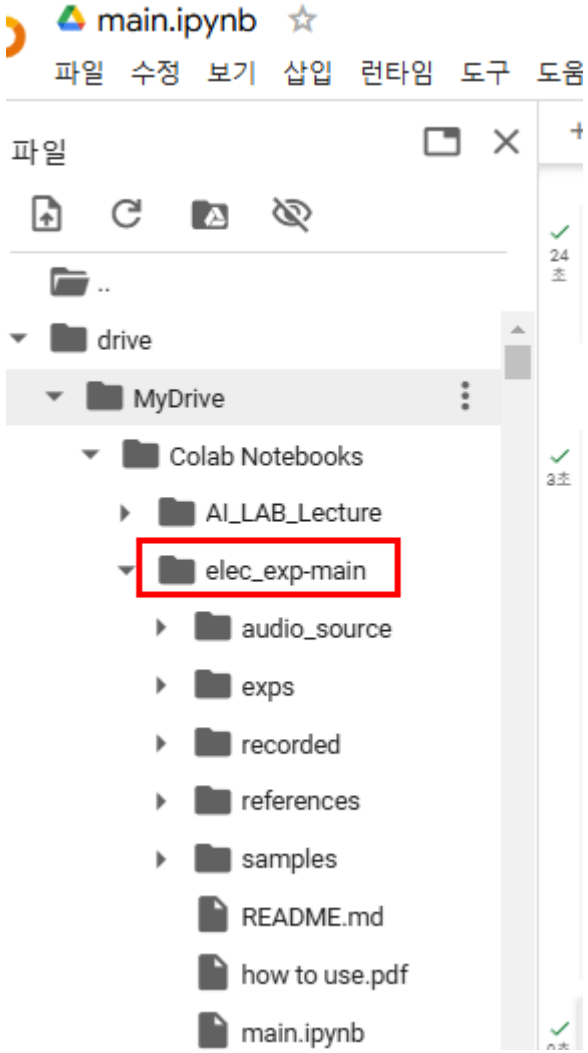
- Experiment analysis




Check the real path

# Term project

- Experiment analysis




```
1 from google.colab import drive
2 drive.mount('/content/drive')
3 colab_basedir = '/content/drive/My Drive/Colab Notebooks/elec_exp-main'
```

```
Mounted at /content/drive
```

colab_basedir must match the left folder path

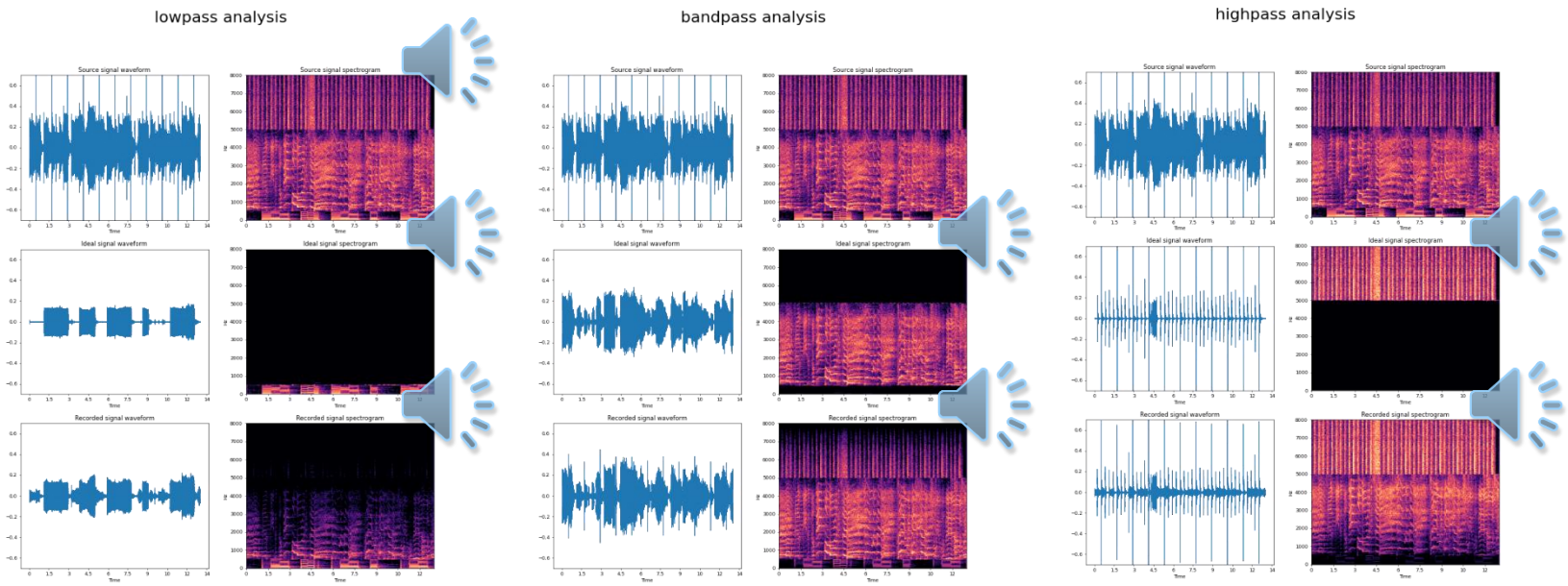
# Term project

- Experiment analysis
  - ✓ In main.py or main.ipynb, modify only the 'sample_idx' and 'filter_mode' variables
    - 'sample_idx' is the index of the audio sample: enter an integer from 1 to 5
    - Set 'filter_mode' to 'lowpass', 'highpass', or 'bandpass' for the filter you are testing

```
[4]  # MODIFY THE SAMPLE AND FILTER
     sample_idx = 1 # 1 ~ 5
     filter_mode = 'lowpass' # lowpass, highpass, or bandpass
```

# Term project

- Experiment analysis
  - ✓ Compare the waveform and spectrogram of the input (source signal), the ideal filter outp ut (ideal signal), and your designed filter output
    - A spectrogram is a time-frequency representation: the x-axis is time, the y-axis is frequency
    - Running the script generates a figure like the ones below
    - Top: source signal, middle: ideal signal, bottom: designed signal



# Term project

- Experiment analysis
  - ✓ Evaluate the similarity of the ideal and designed signals with the scale-invariant signal-to-noise ratio (SI-SNR)
    - An objective measure of the signal relative to the distortion introduced by processing

$$SI-SNR := 10\log_{10}\left(\frac{\|\boldsymbol{S}_{target}\|^2}{\|\boldsymbol{e}_{noise}\|^2}\right),$$

The smaller the denominator (error), the higher the value

$$\boldsymbol{S}_{target} = \frac{\langle \hat{\boldsymbol{S}}, \boldsymbol{S}\rangle}{\|\boldsymbol{S}\|^2}\boldsymbol{S}$$

Ideal signal scaled to the designed signal

$$\boldsymbol{e}_{noise} = \hat{\boldsymbol{S}} - \boldsymbol{S}_{target}$$

Difference between the designed signal and the scaled ideal signal
= filter error

$where,\ S$: ideal signal,

$\hat{S}$: filtered signal (output of your designed filter)

    - The more similar the designed signal is to the ideal signal, the higher the score
      - With a second-order Butterworth digital filter: SI-SNR 7.98 dB
      - An analog filter may score lower than this because of the recording method, etc.

# Thank you

# Q&A

